

Task 1: Scan for Available Wi-Fi Networks

Wi-Fi Networks Found

No.	SSID / Wi-Fi Name	Security
1	[Enter Wi-Fi name]	Password Protected
2	[Enter Wi-Fi name]	Password Protected
3	[Enter Wi-Fi name]	Open / Unsecured
4	[Enter Wi-Fi name]	Password Protected

Task 2: Mobile Hotspot and IP Address

Steps

1. Open Settings on the smartphone.
2. Turn on Mobile Hotspot / Personal Hotspot.
3. Note the hotspot name and password.
4. Open Wi-Fi settings on the laptop.
5. Connect the laptop to the phone's hotspot.
6. Open Command Prompt on Windows.
7. Run:
 - ipconfig

My Laptop's IP Address

Information	Value
IPv4 Address	192.168.43.162
Default Gateway	192.168.43.1
Connection	Mobile Hotspot

Task 3: WPA2 vs WPA3

What is WPA2?

WPA2 (Wi-Fi Protected Access 2) is a wireless security standard used to protect Wi-Fi networks. It commonly uses AES-based encryption and provides authentication between a device and the wireless network.

What is WPA3?

WPA3 (Wi-Fi Protected Access 3) is the newer generation of Wi-Fi security. It provides stronger protection against password-guessing attacks and improves security compared with WPA2.

Two Important Differences

Feature	WPA2	WPA3
Authentication	Commonly uses WPA2-Personal with a pre-shared key (PSK).	Uses SAE (Simultaneous Authentication of Equals) for WPA3-Personal, providing stronger protection against offline password guessing.
Security	Provides strong encryption when configured correctly, but older deployments can be vulnerable to certain attacks.	Provides improved protection and security features designed to make attacks such as offline password guessing harder.

Task 4: Real-World Public Wi-Fi Cyberattack

Case: Fake Wi-Fi Networks at Australian Airports and Flights – 2024

In 2024, Australian authorities investigated a case involving a man accused of creating fake public Wi-Fi networks at airports and on domestic flights. The Australian Federal Police (AFP) alleged that he created "evil twin" Wi-Fi networks that imitated legitimate free Wi-Fi services. The

investigation began after airline employees noticed a suspicious Wi-Fi network during a flight.

How the Attack Happened

The attacker allegedly used a portable wireless device to create Wi-Fi networks that looked like legitimate public networks.

When people connected to the fake networks, they were reportedly directed to fraudulent login pages that requested information such as email or social-media credentials. The submitted information was allegedly stored on the attacker's devices. Investigators identified activity associated with airports in Perth, Melbourne and Adelaide, as well as domestic flights.

Type of Attack

This is known as an Evil Twin attack.

